

FLOOD FORECASTING AND EARLY WARNING SYSTEM

Forecasting system is essential to predict the likely increase in water level in rivers.

Central Water Commission (CWC) provides statistical methodology (gauge to gauge correlation) based short range flood forecast. Presently, flood forecasts are issued by CWC at 332 stations (133 Inflow Forecast Stations and 199 Level Forecast Stations). Annually, about 10,000 flood forecasts are issued by CWC.

CWC is currently providing a near real time five-day advisory flood forecast on its web portal <https://aff.india-water.gov.in/> for 20 major river basins of the country. The five day advance forecast is generated using various available rainfall data products like forecast rainfall data GFS (Global Forecast System) and WRF (Weather Research and Forecasting) provided by IMD (Indian Meteorological Department), rainfall estimates namely GsMaP (Global Satellite Mapping of Precipitation) & GPM (Global Precipitation Measurement), as a major input into the model system.

The system is developed in-house using modeling software for flood forecasting which is updated every three hours for all the stations simultaneously in automatic mode during monsoon season.

CWC has also taken up the work of development of near real time Inundation Forecast for Ganga Basin through consultancy under National Hydrology Project (NHP) using High Resolution Digital Elevation Model (DEM) received from NRSC/ Survey of India / State Govt.

The data collected from field stations is transmitted from the site to the concerned Flood Forecasting Center of CWC through wireless and/or telephone/mobile and satellite based telemetry system & VSAT. The Central Water Commission maintains Wireless Stations for near real-time data communication.

These wireless sets work on pre-fixed schedules for receiving the vital hydro-meteorological data immediately after its observation. Now under the modernization program, a satellite based Telemetry System has been installed at various stations for sensor based automatic data collection and satellite based communication.

The data is transmitted to the Earth Receiving Station through Satellite and then to various Modeling Centers through VSAT Systems. The nodal officer of the dam/reservoir shares reservoir related data with CWC through uploading on Water Information Management System (WIMS) or sending through



email/SMS/Phone/Wireless etc.

Dissemination of flood forecasts has also been modernized by having a dedicated website <https://ffs.india-water.gov.in>. In addition, telephone/mobile phone, fax and internet are used for dissemination of flood forecasts to user agencies. Daily Flood Situation Reports cum Advisories are shared with all stakeholders as well as the general public using social media platforms.

This information was given by the Minister of State for Jal Shakti, Shri Bishweswar Tudu in a written reply in Rajya Sabha.

This tool is designed to provide a practical, step-by-step guide to developing and operating the total flood warning system to agencies responsible for creating and communicating flood warnings. It covers predicting flood levels and the likely impacts of a flood, designing and disseminating warning messages, as well as the means of reviewing the system's effectiveness following an event.

It has been prepared to guide flood managers and can be used before, during and after emergencies. It is intended to help decision-makers establish an effective overview of the situation and find answers to their questions quickly. Many countries have already begun to incorporate flood forecasting and early warning in local and national emergency planning systems, and this tool is meant to provide information on developing flood forecasting and early warning schemes to help with this.

In particular, the tool aims to: (i) provide basic information about flood forecasting and early warning, focusing specifically on riverine floods; (ii) identify flood forecasting and early warning components necessary to address flood risks; (iii) provide perspectives on flood forecasting and early warning strategies that are important for planning flood management activities; and (iv) provide guidance for involving local communities and individuals in flood forecasting and early warning.

Current Issue: As the 2022 monsoon season spreads across India, triggering devastating floods in Assam and other parts of northeast India, and with climate change exacerbating extreme weather events, we take stock of India's early warning systems in preventing loss of lives, property, crops and infrastructure.

The Indian government says it has modern, sophisticated early warning systems for floods and cyclones. But the lack of impact-based forecasts that identify risks, poor dissemination of information to people, lack of scientific data on the effectiveness of warning systems and lack of localized action plans to follow warnings, are some

issues that plague India's Early Warning Systems (EWS), we found.

India experienced two of the world's 10 most financially devastating climate events in 2021.

Both events, Cyclone Tauktae and Cyclone Yaas, caused financial losses worth more than \$1 billion each, apart from the loss of lives.

Between 2010 and 2021, the number of people dying due to cyclonic storms has increased, and floods and heavy rains have killed around 1,000 every year from 2013 onwards.

We wrote to the Ministry of Earth Sciences (MoES) for the reasons behind this increase in mortality, especially with early warning systems. The story will be updated when they respond.

Why early flood warning systems do not work

Floods can be riverine, such as the ones seen frequently in Assam or Bihar, or they can be urban floods largely caused by extremely heavy rainfall coupled with poor stormwater drainage systems. In India, heavy rainfall that causes urban floods is monitored by the India Meteorological Department (IMD), while rising water levels in rivers are monitored by the Central Water Commission (CWC).

Presently, they are operated 1,600 hydro meteorological sites by the CWC across the country, covering 20 river basins. Many of these stations are used as flood monitoring stations for formulating flood forecasts.

Flood forecasting comprises level forecasting and inflow forecasting. Level forecasts help the disaster management agencies in deciding mitigating measures like evacuation, shifting people and their movable property to safer locations. Inflow forecasting is used by various dam authorities to optimize the operation of reservoirs and ensure safe passage of floodwaters downstream. It also helps ensure adequate storage in the reservoirs for meeting the demand of water during the non-monsoon period.

Flood forecasts are issued by CWC at 325 stations (132 Inflow Forecast Stations + 199 Level Forecast Stations), as per a standard operating procedure, and it annually issues over 10,000 flood forecasts. CWC has tied up with Google for dissemination of alerts regarding inundation.

For urban floods, among other things, IMD has a doppler weather radar network of



33 stations to support monitoring and forecast of severe weather, such as thunderstorms and cyclones.

The IMD also operates Flood Meteorological Offices(FMOs) at 14 locations (Agra, Ahmedabad, Asansol, Bhubaneshwar, Bengaluru, Chennai, Guwahati, Hyderabad, Jalpaiguri, Lucknow, New Delhi, Patna, Srinagar and Thiruvananthapuram).

But flash floods also pose a challenge to planning agencies, as they cause widespread destruction in a short span. A 2021 NITI Aayog report recommended a “focus on scientific research in development of a model-based system to forecast flash flood (sic) with sufficient lead time”.

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